2.6 Derivatives of e? and ln x — teacher key

Question: What is the slope of e? at every point, and of ln x?

Outline and pacing (42 min)

Tab	Priority	Minutes	If behind
start	1	2	never trimmed
warmup	1	4	never trimmed
predict	1	3	never trimmed
explore	3	8	skipped first
build	2	8	shrinks to 2 of 4
practice	1	8	never trimmed
check	1	4	never trimmed
Huddle h1	clock	minute 18	pulled in by the clock

Targets

- I can state the derivatives of e? and ln x.
- I can differentiate sums that use both rules.
- I can find slopes, tangent lines and normal lines.

Predictions

- The curve $y = e^x$ climbs faster and faster. As x increases, its slopes will... **keep growing.** The curve gets steeper, so its slopes keep growing.
- The curve $y = \ln x$ rises but flattens out. As x increases, its slopes will... **shrink toward 0.** A flattening curve has slopes that shrink toward 0 (they are $1/x$).

Huddle script (answers in green)

Definitions

- **e?:** Its slope at any point equals its **height**: d/dx e? = **e?**.
- **ln x:** Its slope at x is the **reciprocal** of x: d/dx ln x = **1/x**.
- **Horizontal tangent:** The slope is **zero**. The curve e? never has one.

Worked example

Find the slope of f(x) = ln x + $3x^2$ at x = 1.

- f?(x) = 1/x + **6x**
- f?(1) = 1 + **6**
- So the slope is **7**.

Label the curve e?

1. The **point** where we measure.
2. The **tangent line**: its slope equals the height.
3. The curve is **y = e?**.

Every level, with answers

Differentiate

Level 1. Differentiate.
Answer: e^x $\dfrac{d}{dx}e^x = e^x$.

Level 2. Differentiate.
Answer: $\frac{2}{x}$ $2 \cdot \dfrac{1}{x} = \dfrac{2}{x}$.

Level 3. Differentiate term by term.
Answer: $10x + 2e^x - \frac{5}{x}$ $10x + 2e^x - \dfrac{5}{x}$.

Level 4. Differentiate. Rewrite the root as a power if that helps.
Answer: $\frac{6}{2\sqrt{x}} - \frac{6}{x} + e^x$ $\dfrac{6}{2\sqrt{x}} - \dfrac{6}{x} + e^x$.

Slope at a point

Level 1. Find the slope of $y = e^x$ at $x = 0$.
Answer: 1 The slope equals the height, $e^0 = 1$.

Level 2. Let $f(x) = 6e^x + 5x$. Find $f'(0)$.
Answer: 11 $f'(x) = 6e^x + 5$, so $f'(0) = 6 + 5 = 11$.

Level 3. Let $f(x) = 6\ln x + x^2$. Find $f'(4)$.
Answer: 9.5 $f'(x) = \dfrac{6}{x} + 2x$, so $f'(4) = 9.5$.

Level 4. Let $f(x) = 6\ln x - \dfrac{6}{x}$. Find $f'(2)$.
Answer: 4.5 $f'(x) = \dfrac{6}{x} + \dfrac{6}{x^2}$, so $f'(2) = 4.5$.

Tangent and normal lines

Level 1. Find the tangent line to $y = \ln x$ at $x = 1$.
Answer: $x - 1$ Point $(1, 0)$, slope 1, so $y = x - 1$.

Level 2. Find the tangent line to $y = 2e^x$ at $x = 0$.
Answer: $2 + 2x$ Point $(0, 2)$, slope 2, so $y = 2 + 2x$.

Level 3. Find the tangent line to $y = e^x - 2x$ at $x = 0$.
Answer: $1 + (-1)\,x$ Point $(0, 1)$, slope -1: $y = 1 + (-1)x$.

Level 4. Find the normal line to $y = \ln x + 4x^2$ at $x = 1$.
Answer: $4 - \frac{x-1}{9}$ Tangent slope 9, so the normal slope is $-\frac{1}{9}$: $y = 4 - \dfrac{x-1}{9}$.

Limits that are derivatives

Level 1. Evaluate $\lim_{h\to 0} \dfrac{\ln(2+h) - \ln 2}{h}$.

Answer: 0.5 It is $f'(2)$ for $f = \ln x$: $\dfrac{1}{2}$.

Level 2. Evaluate $\lim_{h\to 0} \dfrac{e^h - 1}{h}$.

Answer: 1 It is $f'(0)$ for $f = e^x$: $e^0 = 1$.

Level 3. Evaluate $\lim_{x\to 5} \dfrac{\ln x - \ln 5}{x - 5}$.

Answer: 0.2 It is $f'(5)$ for $f = \ln x$: $\dfrac{1}{5}$.

Level 4. Evaluate $\lim_{x\to 0} \dfrac{6(e^x - 1)}{x}$.

Answer: 6 $f(x) = 6e^x$, $f(0) = 6$, so the limit is $f'(0) = 6$.

Horizontal tangents

Level 1. Let $f(x) = 2\ln x - x$ for $x > 0$. At which x is the tangent horizontal?

Answer: 2 $f'(x) = \dfrac{2}{x} - 1 = 0$ gives $x = 2$.

Level 2. Let $f(x) = 10\ln x - 2x$ for $x > 0$. At which x is the tangent horizontal?

Answer: 5 $\dfrac{10}{x} - 2 = 0$ gives $x = 5$.

Level 3. Let $f(x) = 6\ln x - 3x + 7$ for $x > 0$. At which x is the tangent horizontal?

Answer: 2 $f'(x) = \dfrac{6}{x} - 3 = 0$ gives $x = 2$.

Level 4. Let $f(x) = \ln x + \dfrac{7}{x}$ for $x > 0$. At which x is the tangent horizontal?

Answer: 7 $f'(x) = \dfrac{1}{x} - \dfrac{7}{x^2} = 0$ gives $x = 7$.

Common misconceptions

Students say	Say instead
The derivative of e? is x·e?$^{-1}$.	The power rule is for x?. In e? the variable is the exponent, and e? is its own derivative.
The derivative of ln x is ln x, or 1/ln x.	The slope of ln x at x is the reciprocal of x: 1/x.
The normal line has the same slope as the tangent.	The normal is perpendicular: its slope is –1/m.
y = e? – 4 has a horizontal tangent somewhere.	The slope is e?, which is always positive. There is no x where it is 0.